ECS7032P Assignment 1 – Individual Reflection

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Group report: Decoupled Information-Set MCTS with a Card-Value Model for Sushi Go!

My role. As group leader I organised the project and divided the workload so that each of us carried an equal share: Halit took the code, Weronika most of the research, and I coordinated, joined the research and led our decisions. Early on we agreed to build on BasicMCTS, the permitted base agent, with every modification as a switch, so that one code base supports the ablations. With the group I set the research questions (which adaptations make the agent stronger, and how their value changes with the time per decision and the number of players) and planned the experimental protocol: 500 games per condition, 95% intervals, budgets from 100 to 900 ms, three and four players, and a final validation on deals never used in development. I also kept the report one coherent story rather than three separate projects.

My teammates. Halit was responsible for the code side: the agent package in TAG, the experiments and the competition files. Weronika did most of the research, on information-set MCTS, determinisation, decoupled UCT and max-n, and her findings settled our main technical choices.

Reflection. The most useful lesson was how much evidence a claim needs: with 500 games a win rate still has an interval of about ±4.4 percentage points, so several promising changes turned out to be noise. Planning the held-out validation paid off: our agent won 52.6% of three-player games on unseen deals, within sampling error of our earlier results. Next time I would fix the deal sets earlier, as our first comparisons used different deals and could not be paired.

Use of generative AI: as our report states, we used Claude (Anthropic) for the code and the drafting; this reflection was also drafted with Claude from our own account of the work.

Word count: 294 (without the heading)